C5 Energy Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Exothermic and Endothermic Paper 1

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Exothermic reaction				Transfers energy to the surroundings; temperature rises
2	Endothermic reaction				Takes in energy from the surroundings; temperature falls
3	Examples of exothermic reactions				Combustion, oxidation, neutralisation
4	Examples of endothermic reactions				Thermal decomposition, citric acid + sodium hydrogencarbonate
5	Uses of exothermic reactions				Hand warmers, self-heating cans
6	Use of endothermic reactions				Sports injury cold packs
7	Activation energy				Minimum energy particles need to react
8	Bond breaking: endothermic or exothermic				Endothermic
9	Bond making: endothermic or exothermic				Exothermic
10	Energy change from bond energies				Bonds broken – bonds made

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C5 Energy Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 2: Reaction Profiles and Measuring Energy Paper 1

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Products on an exothermic profile				Lower than reactants
2	Products on an endothermic profile				Higher than reactants
3	Sign of energy change: exothermic				Negative
4	Sign of energy change: endothermic				Positive
5	Conservation of energy				Energy is not created or destroyed
6	Container for measuring temperature changes				Polystyrene cup with lid
7	Why polystyrene is used				Good insulator
8	Why a lid is used				Reduces heat loss
9	Main source of error				Heat loss to surroundings
10	Units of bond energy				kJ/mol

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C5 Energy Changes – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 3: Chemical Cells and Fuel Cells Paper 1 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Simple chemical cell				Two different metals in an electrolyte
2	Factors affecting cell voltage				Metals used and the electrolyte
3	Why non-rechargeable cells stop working				Reactants are used up
4	Rechargeable cell				Reactions reversed by an external current
5	What a hydrogen fuel cell is supplied with				Hydrogen and oxygen (air)
6	Product of a hydrogen fuel cell				Water
7	Overall equation				$2H_2 + O_2 \rightarrow 2H_2O$
8	Half-equation at the negative electrode				$2H_2 \rightarrow 4H^+ + 4e^-$
9	Half-equation at the positive electrode				$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$
10	Disadvantage of hydrogen fuel cells				Hydrogen is hard to store and flammable

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.